

Machine Design Final Report

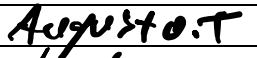
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Table of Contents

Introduction.....	3
Concept Selection Summary	3
Detailed Design.....	4
Testing and Design Evaluation	5
Recommendations and Future Improvements.....	5
Closing Comments	6
High Risk Components and Analysis.....	6
Shaft:	6
Chain sprocket:	7
Belt roller:	7
Deliverables Highlights (b-g)	7
Torque Speed Curve.....	8
Appendix.....	9
1. Stress Analysis.....	9
Shafts.....	9
Belt Roller.....	10
Sprocket	11
Kinematics Analysis	12
Gear box analysis:.....	12
2. Component Drawings	14
3. Assembly Drawings.....	16
4. Bill of Materials.....	16
5. Prototype Photographs and Test Documentation.....	17

Introduction

The goal of this project was to design and fabricate a mechanically driven special-effects machine suitable for film or theatre production. The system was required to rely entirely on mechanical power transmission, with the motor serving as the only electrical component. Additional constraints included a maximum disassembled size of $32" \times 34.5" \times 27.5"$, the use of at least one shaft, and the completion of SolidWorks modelling, stress analysis, and kinematic analysis to ensure that the mechanism could operate safely and reliably before fabrication.

The problem addressed by this project involves creating the illusion of vehicle motion without physically moving a car, an approach commonly used in filmmaking to reduce risk and cost. To achieve this effect, the final design simulates a high-speed highway chase using two synchronized moving belts. The lower belt functions as the roadway beneath a stationary model car, while the upper belt displays a repeating skyline and streetlight pattern to represent the passing environment. When these belts move together, they generate the visual impression of forward motion, allowing a highway chase scene to be filmed safely and consistently.

The completed mechanism uses a pulley-and-shaft system driven by a single motor, with bevel gears transferring power to the vertical background belt. This configuration produces smooth and coordinated belt movement while keeping the system mechanically simple and stable. The remainder of this report outlines the design development process, summarizes the concept selection rationale, presents the detailed engineering analyses, documents prototype testing, and evaluates the final system's performance as a practical special-effects solution.

Concept Selection Summary

The concept developed earlier in the semester was a motion platform that created the illusion of a moving vehicle through coordinated belt movement. In its early stages, the design focused only on a rotating background scenery belt. Although this provided some sense of motion, it did not convincingly represent a highway chase on its own. The concept was expanded by introducing a second moving belt beneath the car, allowing the wheels to spin and the roadway to appear in motion. This addition significantly improved realism and strengthened the overall special-effects effect.

During the concept evaluation process, several alternative approaches were considered. One idea involved keeping the car supported using fishing lines, but this method lacked stability and did not securely position the model during belt movement. Zip ties were later adopted because they provided a firmer and more reliable method of restraint. The team also initially planned to drive the system using a belt-pulley arrangement; however, this was replaced with a chain-sprocket transmission due to the chain's improved grip, reduced slippage, and more consistent speed transfer at higher rotational speeds.

The final concept, a dual-belt system powered through a two-sprocket mechanical reduction, was chosen because it offered the most convincing visual effect while remaining mechanically simple, stable, and feasible within the project constraints. Combining a moving background with a moving roadway created a layered motion effect essential for simulating a highway chase, making this concept the most effective and practical option.

Detailed Design

The final design uses two separate belt systems to create the illusion of a moving vehicle scene. A ground belt under the toy cars and a painted background belt behind them. A wooden frame holds all shafts, drums, bearings, and the DC motor. The overall layout was chosen to keep the mechanism simple, strong and easy to assemble.

Power comes from a small sprocket fixed directly to the motor shaft, which drives a larger sprocket through a chain. We originally planned to use a belt driven system, but calculations showed that the belt would likely slip under torque from motor on such a small pulley. The chain and sprocket option was a better option offering consistent torque transfer and much better durability.

The larger sprocket is press fit to the main shaft, which rotates the shaft and drum controlling the ground belt. Both ends of the shaft sit in bearings fit in the wooden side panels. A second freely rotating shaft with a drum supports the opposite end of the ground belt.

The background belt is driven by a shaft above the ground belt. Power is transferred through a pair of bevel gears, allowing rotation to change direction by 90 degrees without pulleys or alignment issues. This upper shaft also uses a drum with a painted belt wrapped around it to simulate the city skyline moving past the cars.

All dimensions for shaft lengths, sprocket ration, drum diameter, bearing placement were determined using the calculations in our Excel sheets (Appendix 1). Assembly drawings (Appendix 3) show the spacing required to keep the belts aligned and gears properly meshed.

Testing and Design Evaluation

Testing focused on verifying that both belts moved smoothly and that the motion matched what we predicted. We also checked for belt drift, shaft wobble, and gear alignment. During operation, the chain drive consistently delivered enough torque to move the ground belt under the load of the two toy cars. The bevel gears also engaged properly, and the background belt rotated without slipping.

A few issues appeared during testing. Both the ground and background belts moved faster than we wanted which meant we couldn't see the scenery on the background. To bring the speed down we added friction pads on the driven shaft and the driven background scenery shaft. This worked and we were able to half the initial speed.

Despite this issue, the system met the main project constraints. Both belts moved reliably, the motor operated within safe torque levels, and the scene successfully created the illusion of motion. The main safety considerations involving the rotating chain and sprockets as well as the bevel gears were covered with aluminum sheet metal to prevent fingers from getting caught during operation.

Recommendations and Future Improvements

For the future, making an adjustable tensioning mechanism for the belts would help make the motion smoother. Using CNC machined parts instead of hand cut wooden frame would significantly improve accuracy and long-term durability. Adding an adjustable tensioning system for the chain would also help with long term reliability, since it would allow quick fine tuning as components wear.

Another improvement would be making the background scene modular and interchangeable. Instead of one fixed painted belt, the design could use removable scenery belts with different artwork or themes. This would increase versatility and make

the system more practical for real special effects use, allowing a single mechanism to support multiple movie scenes without rebuilding the entire device.

Adding a simple motor speed controller would also allow better control between the ground and background motion, improving the realism of the effect and reducing stress on the drivetrain.

Closing Comments

The completed mechanism successfully demonstrates the intended special effect of a highway chase scene using a fully mechanical system. The synchronized motion of the roadway belt and rotating background creates a convincing illusion of forward movement while keeping the vehicle stationary, meeting the safety and cost-effectiveness goals defined at the start of the project. The final prototype also operates smoothly using a single motor, with the pulley system, shafts, and bevel gears providing reliable and consistent power transmission.

Testing showed that the design functions as expected, although several areas for improvement were identified. Enhanced belt tensioning, refined material finishes, and a more enclosed structure would increase durability and visual quality, especially if the system were to be used in a professional production setting. Small adjustments to alignment and long-term wear resistance could further improve performance during extended use.

Overall, the project demonstrates a clear and functional solution for producing mechanical special effects in film or theatre environments. The final design aligns with the engineering analysis completed throughout the semester and provides a solid foundation for future refinement or scale-up. The mechanism offers a practical and accessible method for creating dynamic highway chase visuals, showing strong potential for integration into real production scenarios.

High Risk Components and Analysis

Shaft:

The shaft connects the motor to both the main conveyor drive sprocket and the bevel gear that powers the moving background.

- **Analysis:** Bending and torsion stresses using maximum torque from the motor and belt tension.
- **Material:** AISI 1030 Cold Rolled steel, chosen for its higher yield strength (~440 MPa), dimensional stability, and machinability.
- **Risk Mitigation:** Using a conservative factor of safety (>2.0), and machining keyways with tight tolerances to prevent slip or misalignment.

Chain sprocket:

The team determined that a chain and sprocket system offers reliable torque transfer with no slippage, making it suitable for maintaining consistent roller speed. Although it produces more noise than a belt system, it provides stronger positive engagement and handles higher loads effectively.

- **Analysis:** Chain tension, sprocket tooth loading, shear on chain pins, and torque transfer through mechanical engagement.
- **Risk Mitigation:** Maintain proper chain tension, ensure accurate sprocket alignment, lubricate the chain to reduce wear, and verify that both components can withstand repeated cyclic loading.

Belt roller:

The belt transfers motion from the drive pulley to the driven roller. As it operates under continuous loading, tension control and wear resistance are key considerations.

- **Analysis:** Belt tension, contact friction, and roller bearing loads under steady operation.
- **Risk Mitigation:** Belt selected with a safety factor of three over the design load.
Alignment tolerance of ± 0.5 mm maintained during assembly to prevent edge wear.

Deliverables Highlights (b-g)

- **(b) Shear Analysis:** Excel sheets will be used to evaluate torsional, bending and shear stress for the shaft, pulley, and belt tension.
- **(c) Kinematic Analysis:** Calculations confirm a pulley ratio producing a conveyor belt speed of 0.486 m/s, achieving smooth motion.

- **(d) Assembly Model:** Full SolidWorks assembly with final design, tolerances, and clearances completed.
- **(e) Engineering Drawings:** Detailed part and assembly drawings created.
- **(f) Bill of Materials:** Includes shafts, sprockets, chain, bearings, gears, pulley, belt, wood frame and motor.
- **(g) Cost Information:** Funding approved. All suppliers verified for part availability.

Torque Speed Curve

To create the torque speed curve for the given motor, the speed of the motor at different torques needed to be measured. To measure the speed, a pulley was attached to the motor, then a string was tied around the pulley, and different weights were tied to the end of the string. The torque could be found by using a scale to measure the mass of the weights in kilograms, then the mass is multiplied by gravity (9.81m/s^2) and the radius of the pulley (5cm). To find the speed a measuring tape is parallel to the string and a video is then taken when the motor starts of the weight moving straight up as the pulley rotates. This video is then used to find the linear speed by dividing the distance the weight moved by the time it took to travel that distance. To find the speed in rotations per minute, the linear speed is converted to meters per minute by multiplying by sixty, then it's divided by the circumference of the pulley ($C=2\pi r$).

Table 1: Torque-Speed data

Torque (oz-in)	Speed (rpm)
0	5000
27.29	4800
44.11	4285.7
52.41	4000
79.61	3692.3
327.12	0

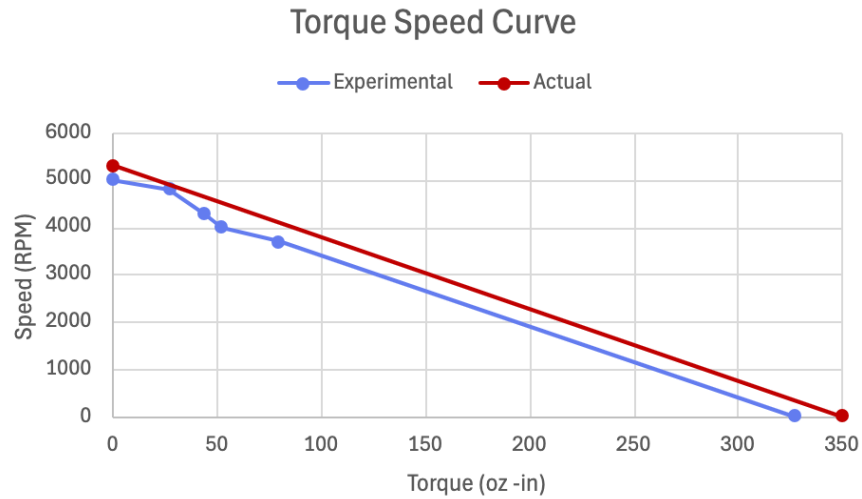


Figure 1: Torque-Speed Curve

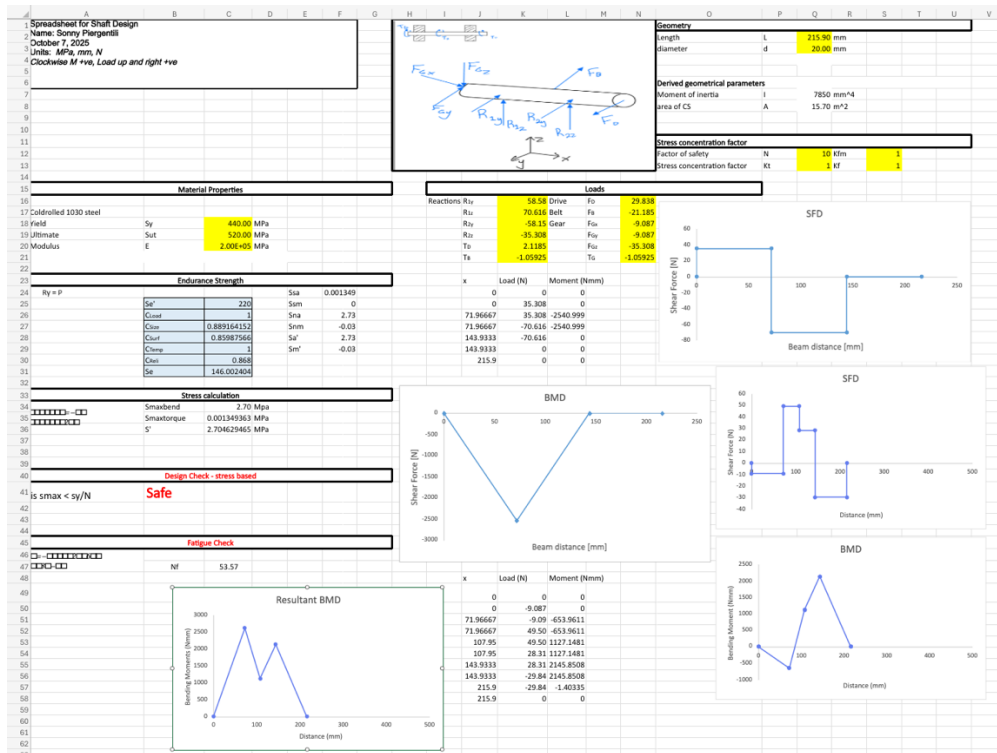
Appendix

1. Stress Analysis

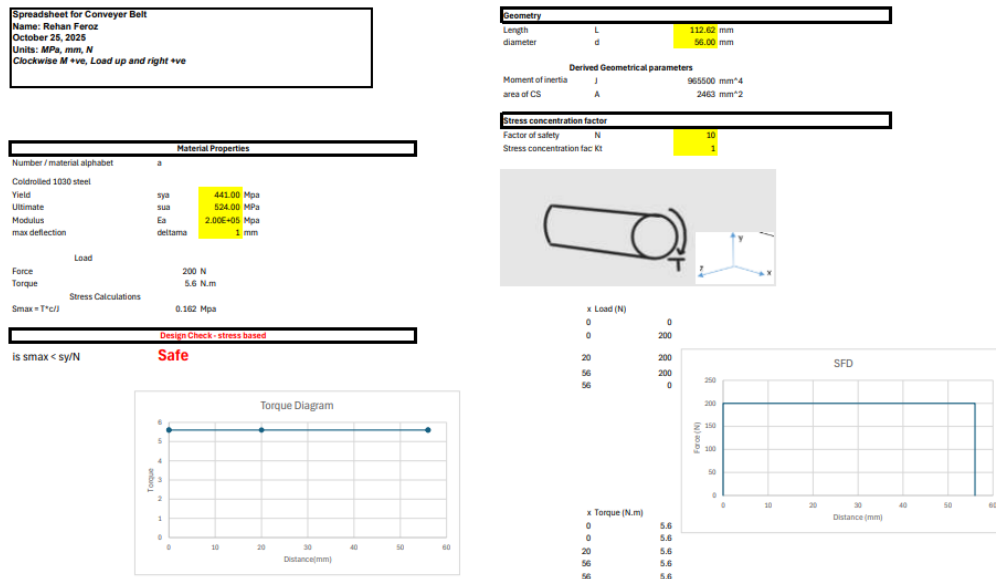
Shafts

Equations used to solve for Reaction Forces acting on shaft	
From chain drive, pulley and tangential bevel gear component	$F = \frac{\text{Torque}}{\text{radius}}$
Axial and radial bevel component	$F_r = F_t \tan \phi \cos \alpha, F_a = F_t \tan \beta \sin \alpha$

Note: angles and radii were found on the website parts were purchased from



Belt Roller



Sprocket

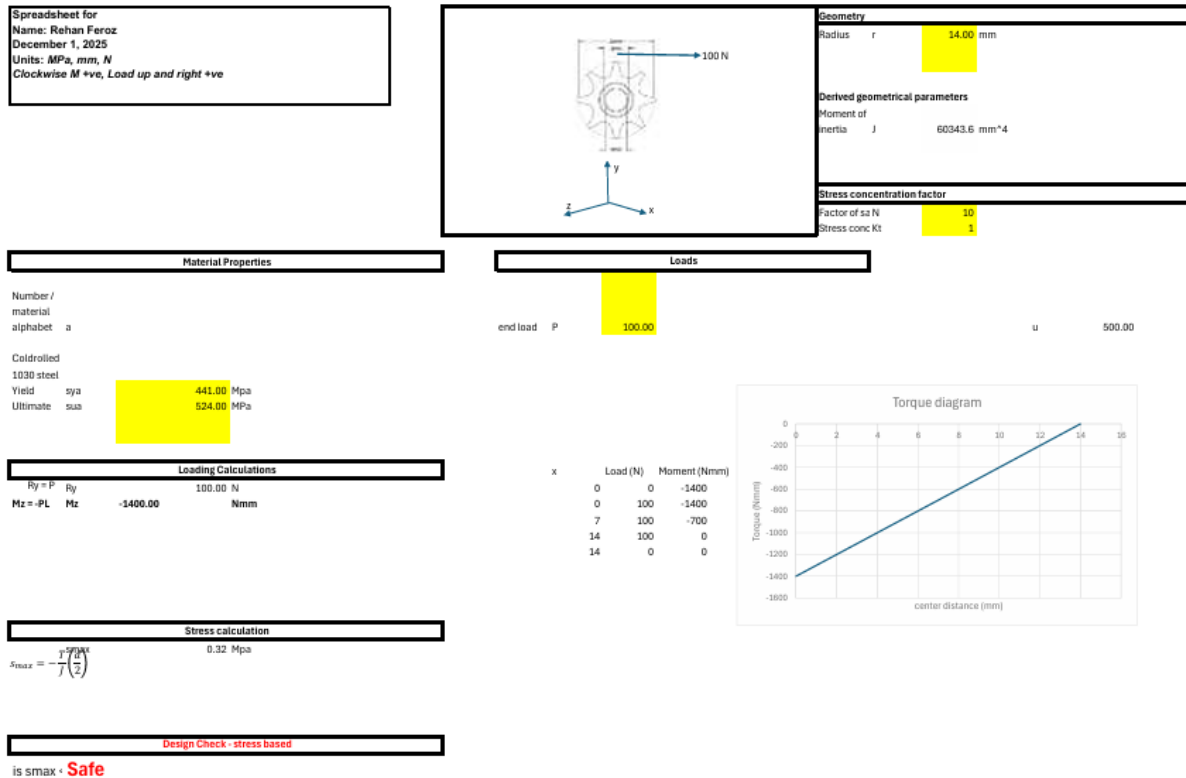


Figure 4: Pulley Excel sheet

Bearing Calculations	
$C = 3050lb, C_o = 1450lb$	$F_R = 20.626lb, F_a = 2.043lb$ (reaction forces from shaft analysis converted to lb)
$\frac{F_a}{V F_R} = 0.1$ ($V = 1$ since inner race rotates)	$\frac{F_a}{C_o} = 0.0014$ (Figure 11-24 in Norton does not include this value so since the axial force is very small it is assumed negligible) $P = 20.626lb$
$L_{10} = \left(\frac{C}{P} \right)^3 = 3233363$ million revelations	

Kinematics Analysis

The objective of this analysis is to determine the linear velocity of the roller drum, which drives the two moving belts in the mechanism: the background scenery belt and the road surface belt. Both belts rely on the rotational speed transmitted through a two-sprocket chain system powered by the motor.

From the torque–speed curve, the motor operates at an approximate speed of:

$$\omega_{motor} = 5000 \text{ rpm}$$

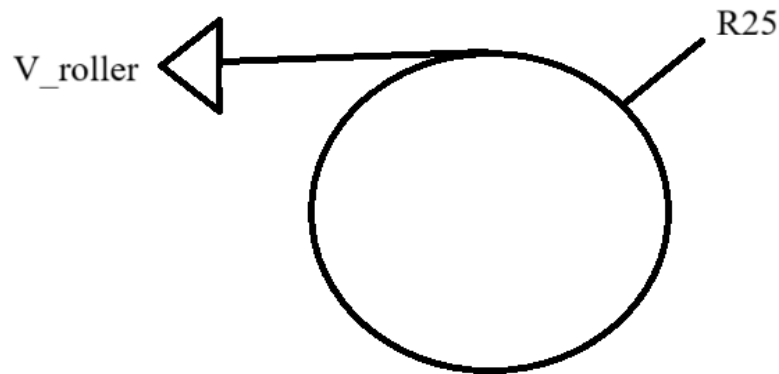


Figure 5: Roller Kinematics FBD

Gear box analysis:

The power transmission consists of two chain sprockets, where Sprocket 1 is mounted on the motor shaft and Sprocket 2 is mounted on the roller shaft.

Given:

$$N_1 = 8, N_2 = 45$$

The speed ratio for a chain drive is:

$$\frac{\omega_{out}}{\omega_{in}} = \frac{N_1}{N_2}$$

Substituting values:

$$\omega_{out} = \omega_{motor} * \frac{N_1}{N_2}$$

$$\omega_{out} = 5000 * \frac{8}{45}$$

$$\omega_{out} = 888.9 \text{ rpm}$$

The sprocket pair therefore provides a reduction from 5000 rpm to approximately 888.9 rpm at the roller shaft.

The output gear (Gear 2) is mounted on the same shaft as a bevel gear, which transmits the rotational motion to the background scenery mechanism.

Linear Velocity of Roller

Given the roller radius:

$$R = 25 \text{ mm} = 0.025 \text{ m}$$

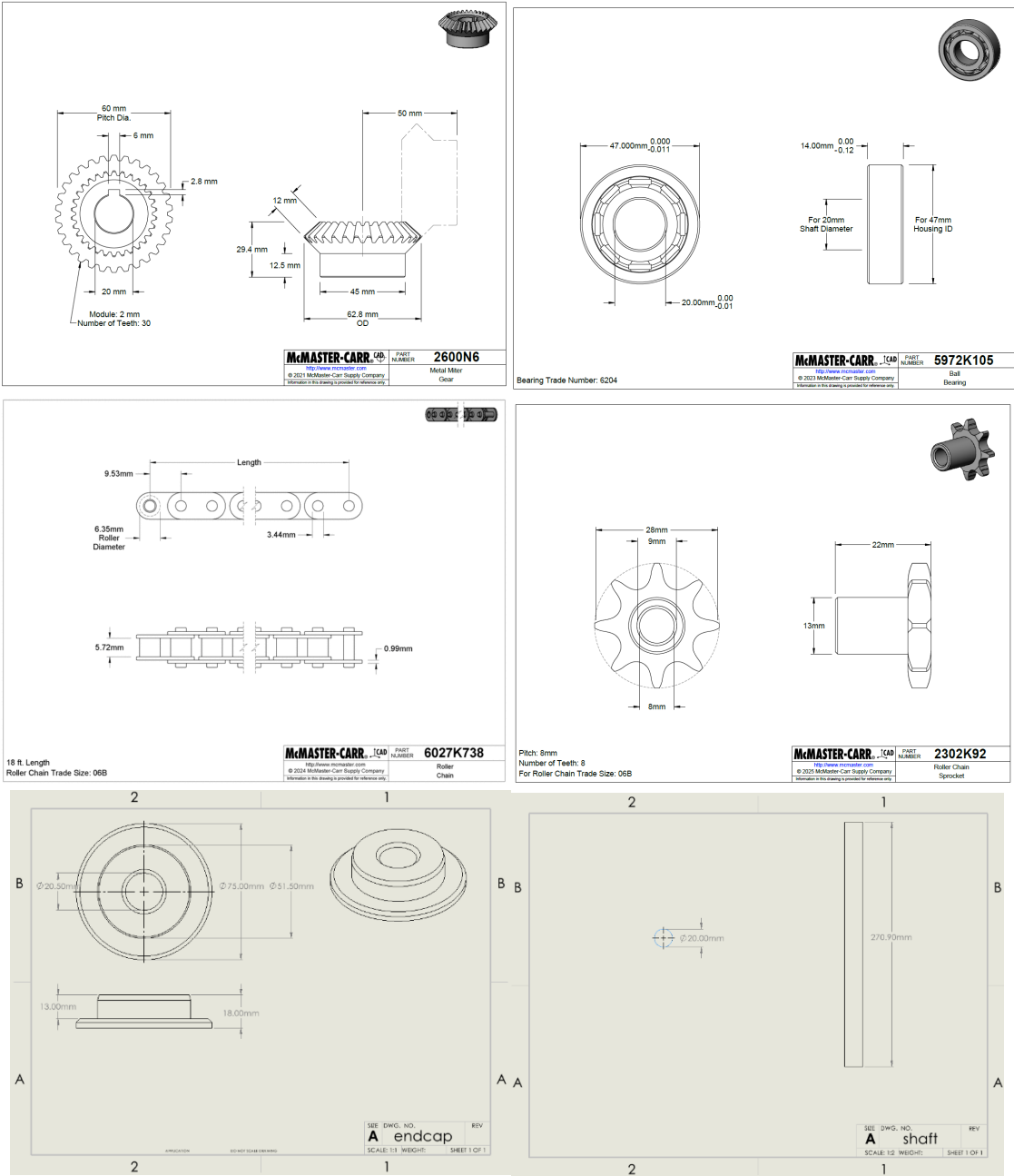
The linear velocity of the roller is:

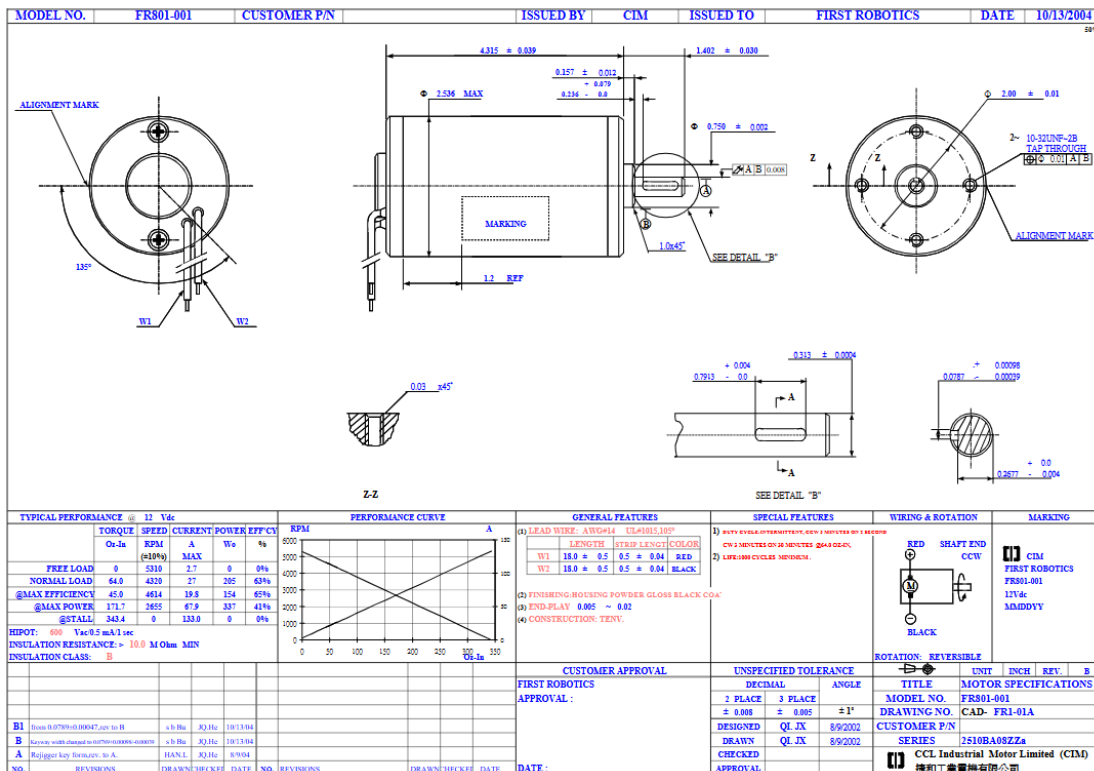
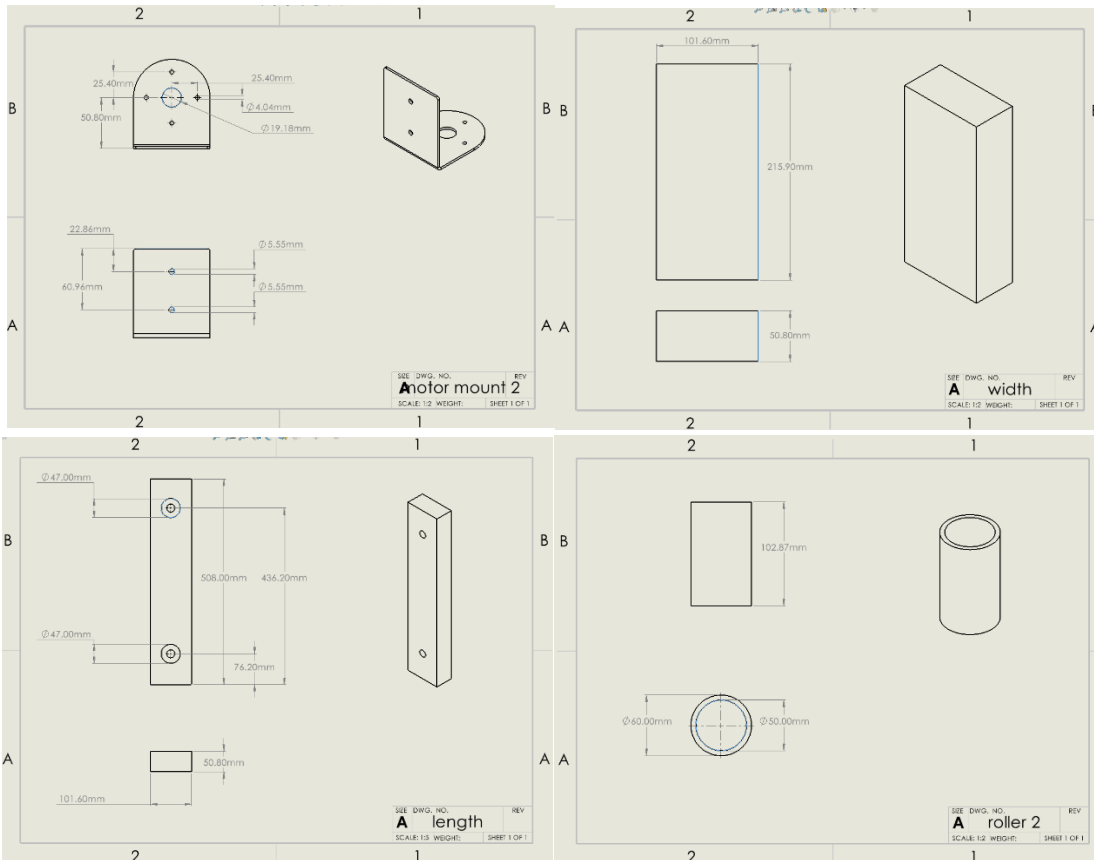
$$V_{roller} = \omega_{out} * R * \frac{2\pi}{60}$$

$$V_{roller} = 93 * 0.025 * \frac{2\pi}{60}$$

$$V_{roller} = 2.33 \text{ m/s}$$

2. Component Drawings





3. Assembly Drawings

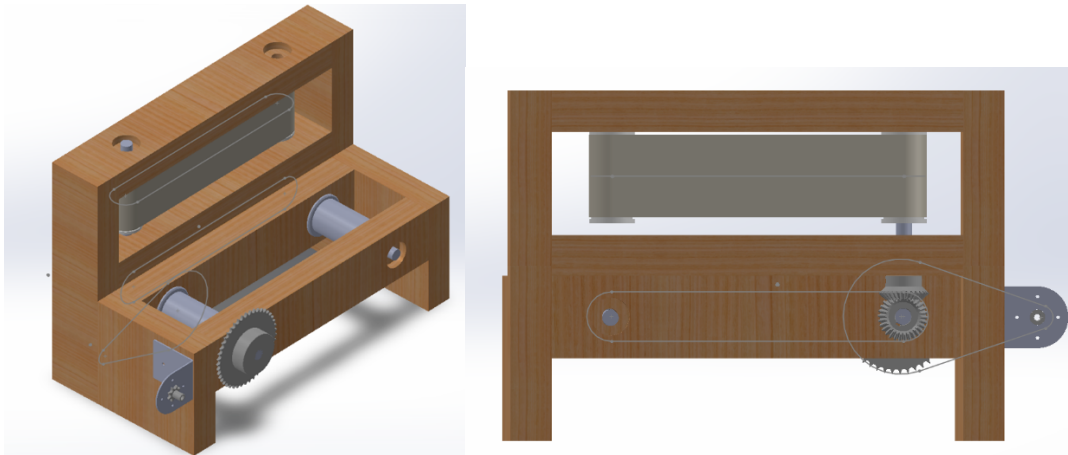


Figure 1: Isometric view and rear view of the device

4. Bill of Materials

Table 2: Bill of Materials tab

Item	Description	Quantity
Wood 2x4 16ft	2-in x 4-in x 16-ft Fir Kiln-Dried Lumber	1
Wood screws	Yellow zinc Interior Wood screws	16
Wood Glue	Gorilla Wood Adhesive	1
4x36 Sanding belt	Gator Sanding Belt Sandpaper	2
1030Cr steel shaft	Carbon Steel	4
Bearings	Ball Bearings	8
Power Source	12V source	1
Motor	12V DC motor	1
Wires	Connection for Power, Switches at motor	1
ABS pipe	Roller Hub	<0.5kg
Chain	Power transmission	22in
Spur Gear	Aluminum	2
Roller Chain	1/2in pitch	22in long
Bevel Gear	38.1mm Diameter	2

Table 3: Cost Information

Item	Cost
Wood 2x4 16ft	\$19.97
Wood screws	\$9.98
Wood Glue	\$4.38
4x36 Sanding belt	2x \$9.99
1030Cr steel shaft	\$59.16
Bearings (x8) 5972K105	\$120.61
Motor	Included
Wires	Included
Switch	Included
PC plastic	Included
Sprockets (x2) 2302K207 and 2302K92	\$147.58
Roller Chain 6027K144	\$ 47.59
Bevel Gear (x2) 2600N6	\$230.83

5. Prototype Photographs and Test Documentation



Figure 18: Pictures of the developmental stages of fabrication



Figure 19: Pictures of mechanism testing

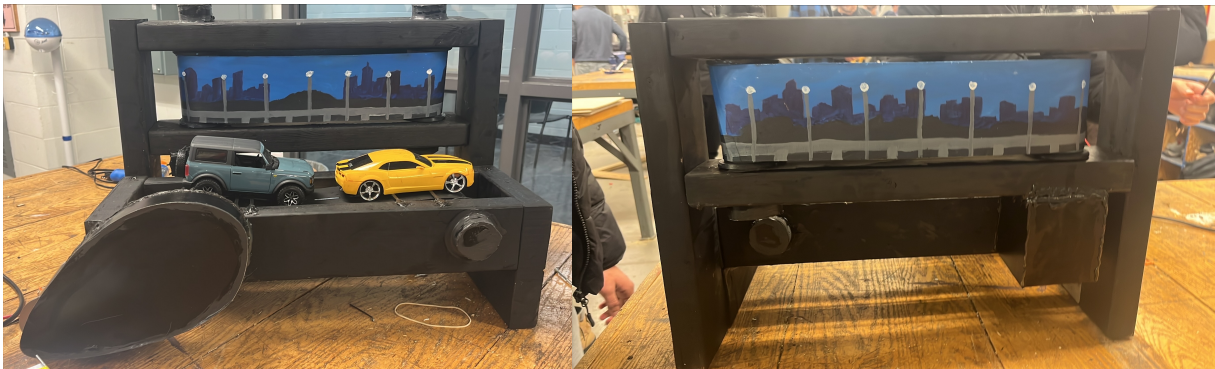


Figure 20: pictures of finished design